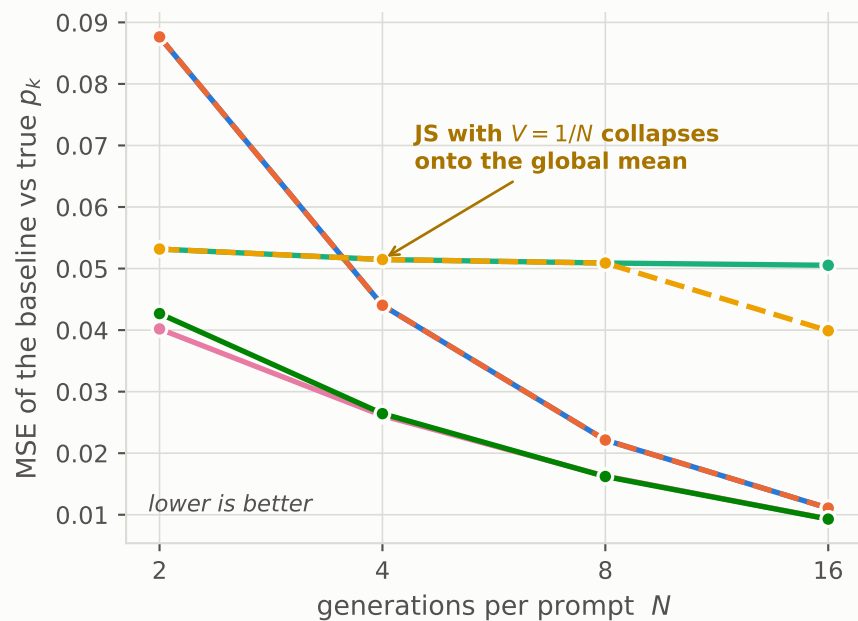
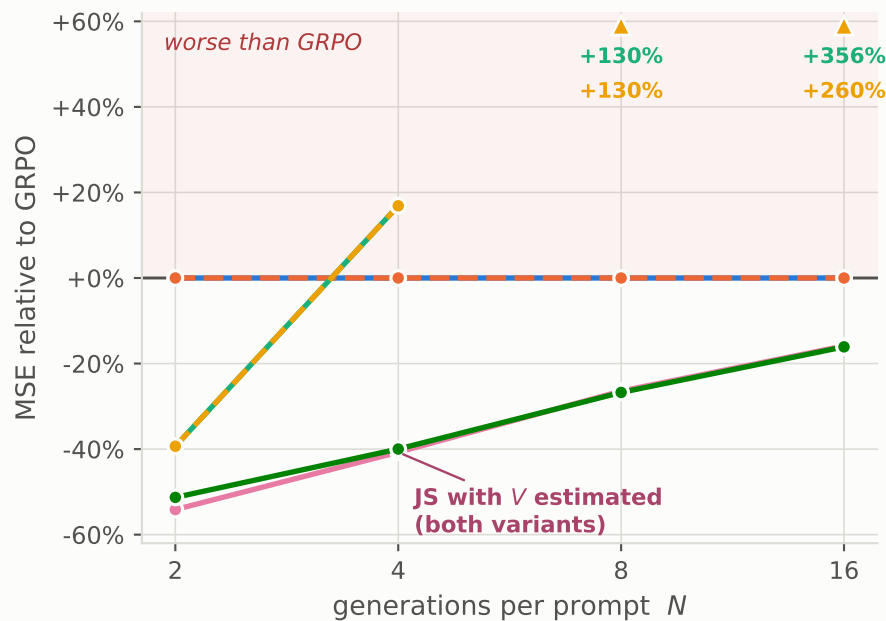


Shrinkage baselines for GRPO — Bernoulli rewards, K=32 prompts, 2000 trials, $p_k \sim \text{Beta}(1.2, 2.2)$

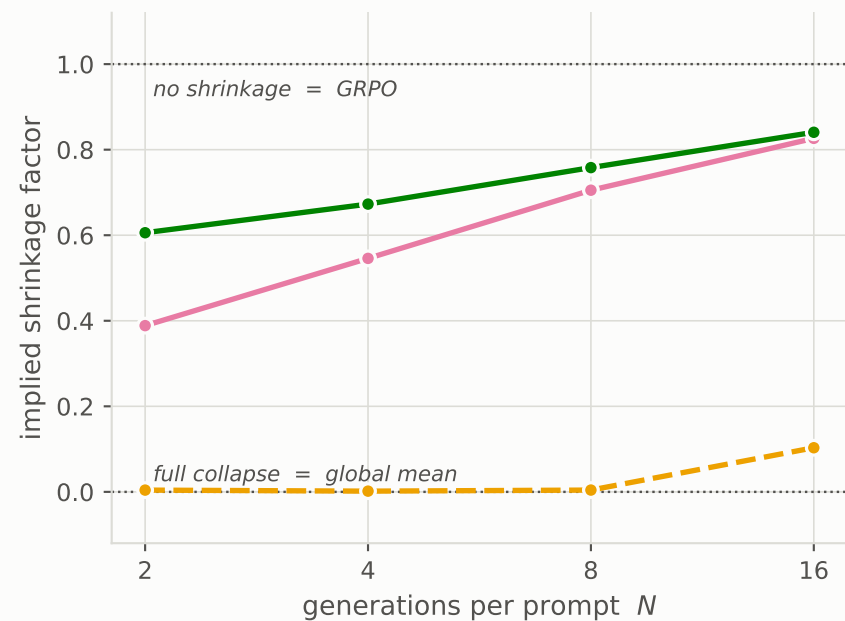
A. Estimation error



B. Change vs plain GRPO



C. Why — how hard each variant shrinks



—●— GRPO (group mean) —●— RLOO (leave-one-out) —●— global mean —●— JS, $V=1/N$ (GRPO.md as written) —●— JS, V estimated —●— JS two-level LOO (arXiv:2511.03710)